

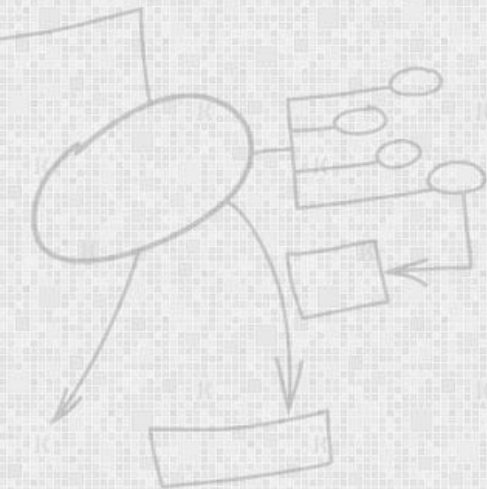
EC-360

SOFTWARE ENGINEERING

Lecture 3

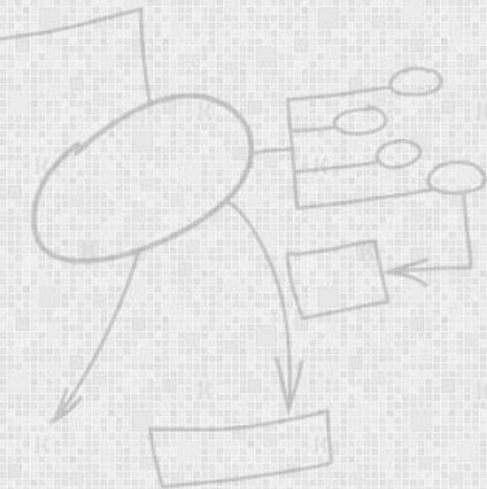
Software Requirements

By
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Today's Lecture: Agenda

- Requirements
- Requirements Classification
- Ways of Writing Requirements



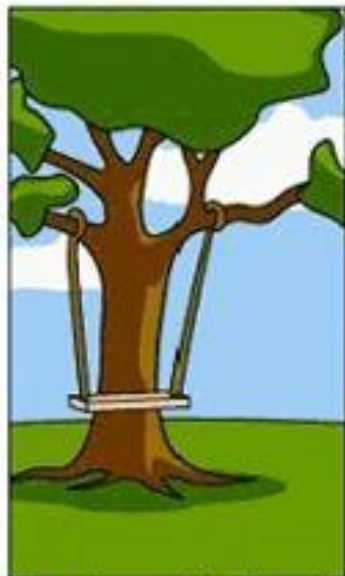
Requirement: Definition

1. A condition or capability needed by a **user** to solve a problem or achieve an objective.
2. A condition or capability that must be met or possessed by a system component to satisfy a **contract, standard, specification**, or other formally imposed document.
3. A documented representation of a condition or capability as in 1 or 2.





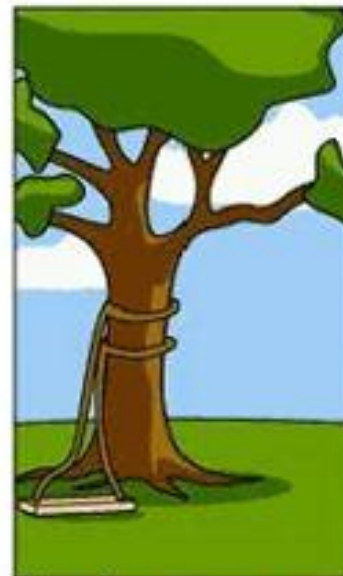
How the customer explained it



How the project leader understood it



How the engineer designed it



How the programmer wrote it



How the sales executive described it



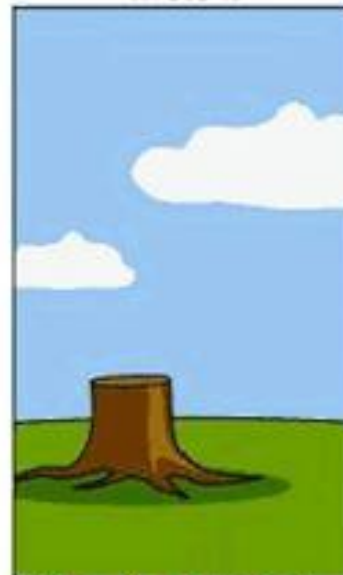
How the project was documented



What operations installed



How the customer was billed



How the helpdesk supported it



What the customer really needed

“FURPS+” Classification of Requirements

- **Functional**
- **Non-Functional (NFRs)**
 - **Usability**
 - The ease with which the system can be learned and operated by the intended users.
 - Help, documentation, required training time, task times for typical tasks, conformance to usability standards, etc.
 - **Reliability**
 - Allowed MTBF (Mean Time Between Failures)
 - Recoverability, etc.

FURPS+ Classification of Requirements

– Performance

- Response time, throughput, ...
- Average, maximum response time for a transaction
- Number of customers/transactions the system can accommodate
- Example:
 - Buyers want to complete sales processing very quickly. One bottleneck is external payment authorization. Our goal: authorization in less than 1 minute, 90% of the time.

– Supportability:

- Supportability is the ability of the software to be easily modified to accommodate enhancements and repairs



FURPS+ Classification Cont.

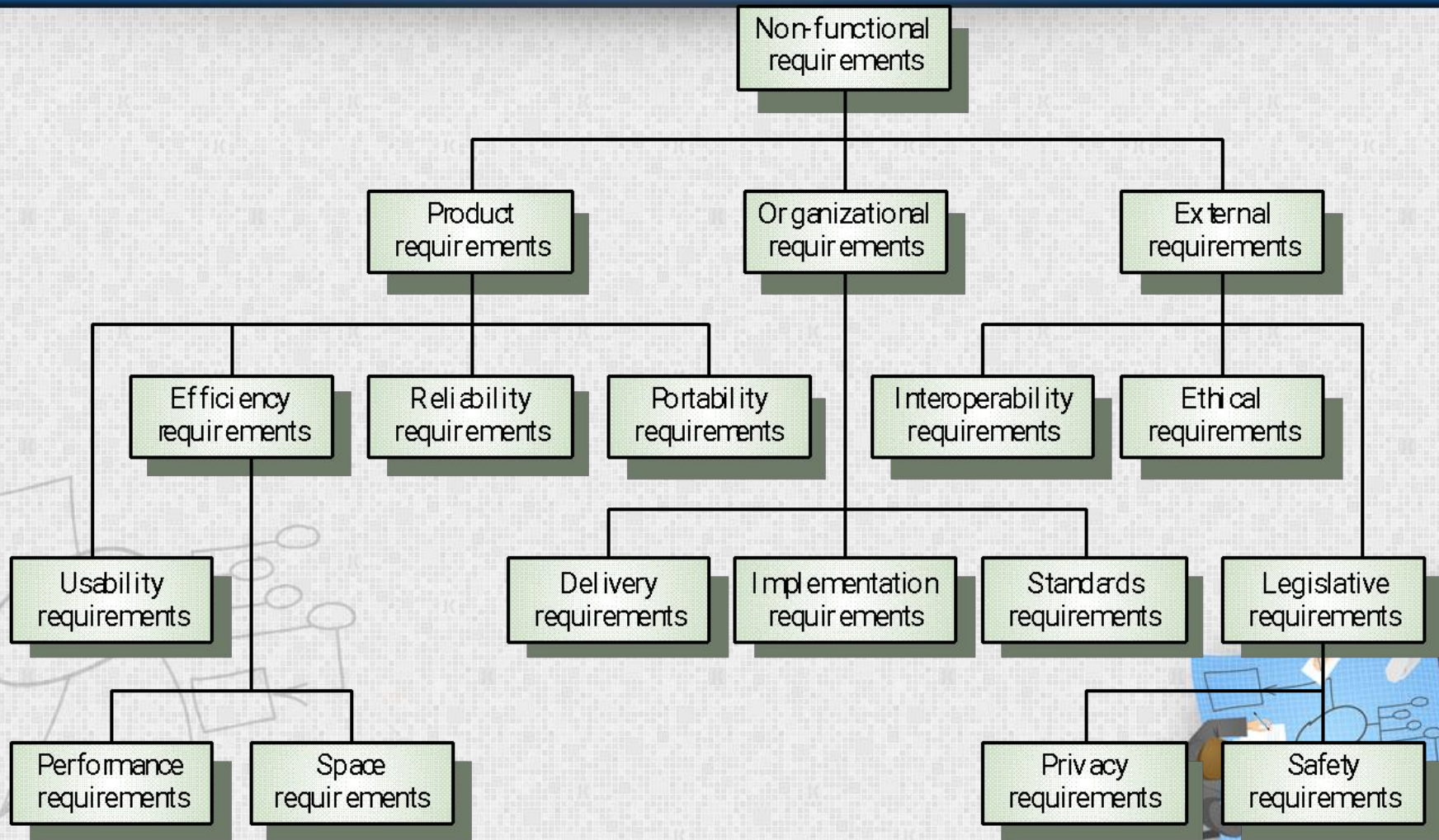
- **+** indicates additional constraints such as:
 - **Design Constraints**
 - E.g. requiring a **relational** database
 - **Interface constraints**
 - i.e. protocols for interaction with external systems
 - **Implementation Constraints**
 - Required platform, implementation language
 - **Physical constraints**
 - Hardware devices



FURPS+ Classification Examples

- "The persistence will be handled by a relational database" **FURPS+ Type?**
 - **“+” design** constraint.
- "The database will be Oracle 8i" **FURPS+ Type?**
 - **“+” implementation** constraint.
- "The system will run seven days a week, twenty-four hours per day" **FURPS+ Type?**
 - **“R” reliability** requirement.

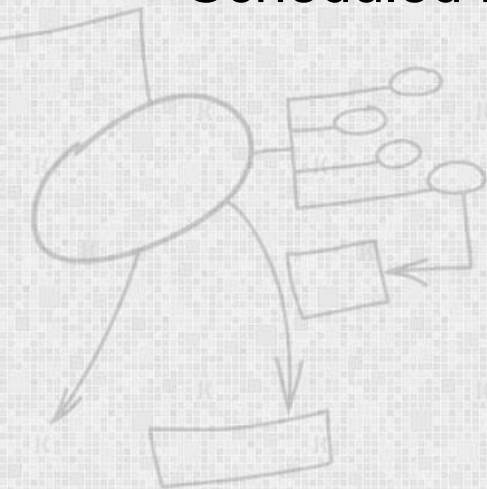
NFR Classification



Non-functional requirements

- **Availability**

- Availability is a measure of the planned *up time* during which the system is actually available for use and fully operational.
- More formally, availability equals the mean time to failure (MTTF) for the system divided by the sum of the MTTF and the mean time to repair the system after a failure is encountered.
- Scheduled maintenance periods also affect availability.



Non-functional requirements

- **Robustness**

- *Robustness*, sometimes called *fault tolerance*, is the degree to which a system continues to function properly when confronted with invalid inputs, defects in connected software or hardware components, or unexpected operating conditions.
- Robust software recovers gracefully from problem situations and is forgiving of user mistakes.
- When eliciting robustness requirements, ask users about error conditions the system might encounter and how the system should react.



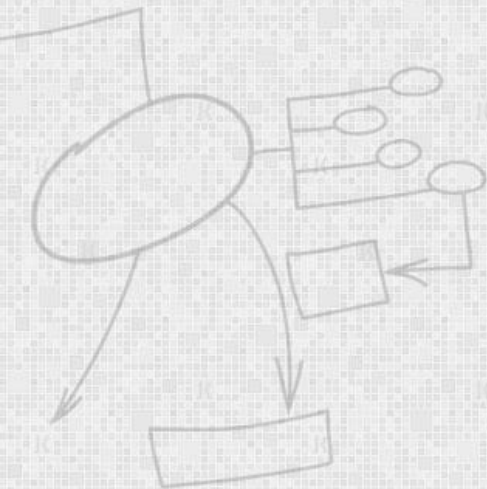
- Integrity

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Writing Requirements

- Plain Natural Language
 - System should be able to generate pay slip on customer's request
 - System should allow user to mark student's attendance
 - As faculty member, user should be able to view student grades in his respective subjects
- Benefits
- Problems

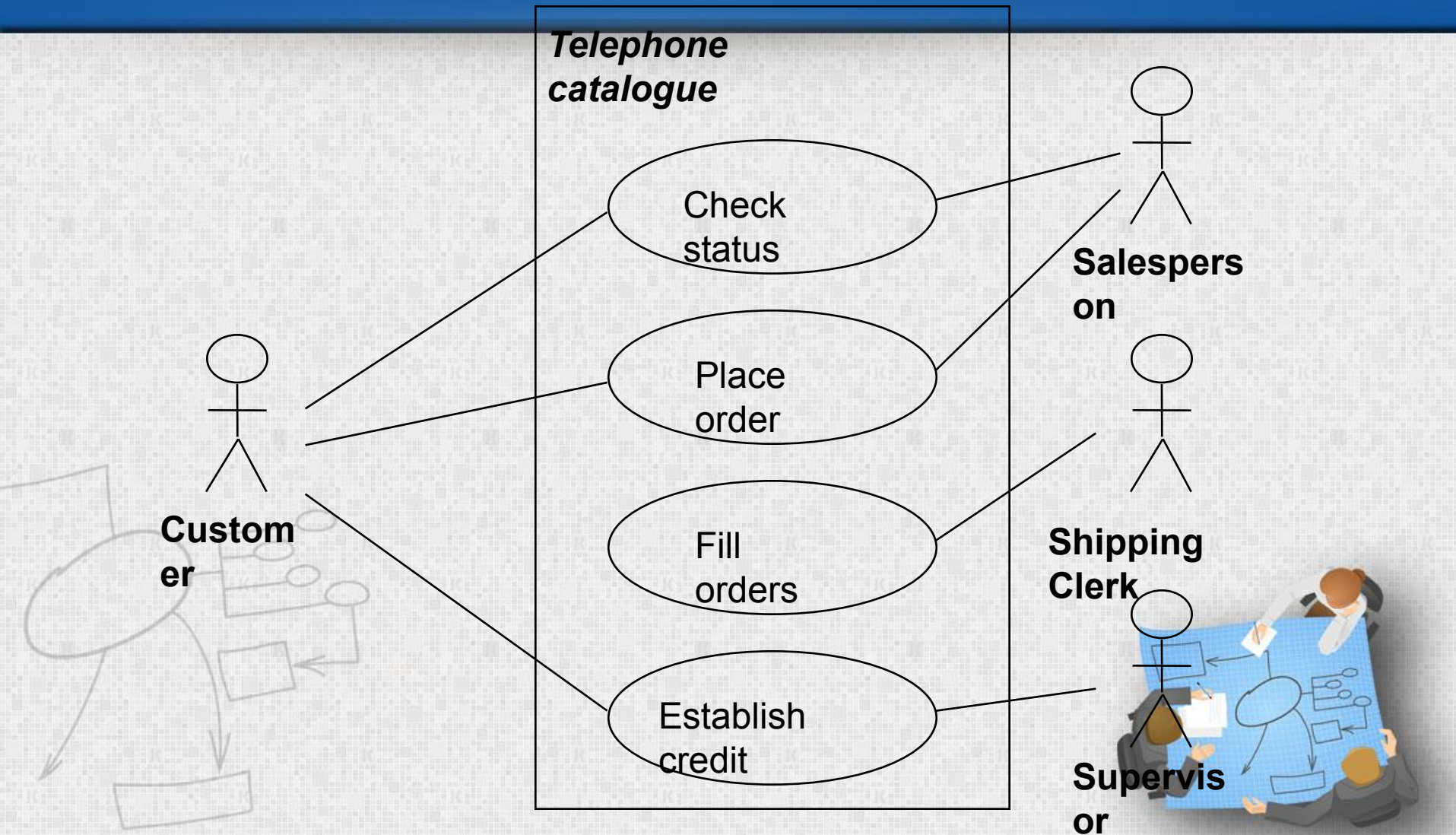


Structured Language (Decision Table)

<i>Condition Stub</i>		<i>Condition Entry</i>					
If	Customer is bookstore	Y	Y	N	N	N	N
	Order size > 6 copies	Y	N	N	N	N	N
	Customer is librarian/individual			Y	Y	Y	Y
	Order size 50 copies or more			Y	N	N	N
	Order size 20 to 49 copies				Y	N	N
	Order size 6 – 19 copies					Y	N
Then	Allow a 25% discount	X					
	Allow a 15% discount			X			
	Allow 10% discount				X		
	Allow a 5 % discount					X	
	Allow 0% discount		X				
	<i>Action Stub</i>	<i>Action Entries</i>					



Diagrams: Example Use Case Diagram



Mathematical Notations

- A book can either be in the stacks, on reserve, or loaned out
- $S \Leftrightarrow \neg(R \vee L), R \Leftrightarrow \neg(S \vee L), L \Leftrightarrow \neg(S \vee R)$

If a book is requested, then it is on the shelf or on reserve

- $Q \Rightarrow (S \vee R)$



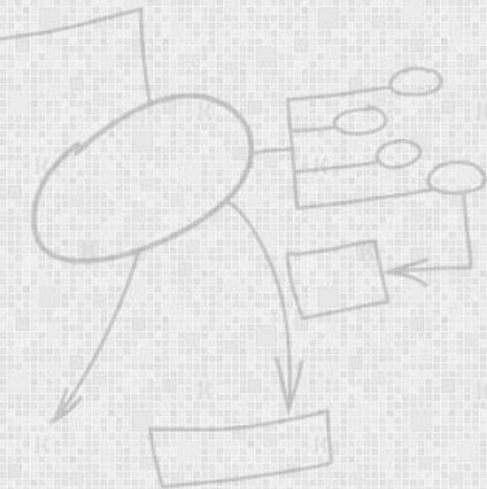
Defining Non-Functional Requirements with Planguage

- To address the problem of ambiguous and incomplete nonfunctional requirements, consultant Tom Gilb (1988; 1997) has developed *Planguage*, a **planning language** with a rich set of keywords that permits precise statements of quality attributes and other project goals.



Defining Non-Functional Requirements with Planguage

- Each requirement receives a **unique tag**, or label.
- The **ambition** states the purpose or objective of the system that leads to this requirement.
- **Scale** defines the units of measurement
- **Meter** describes how to make the measurements.



Non-Functional Requirements with Planguage Database Query

- **TAG** Performance.QueryResponseTime
- **AMBITION** Fast response time to database queries.
- **SCALE** Seconds of elapsed time between pressing the Enter key or clicking OK to submit a query and the beginning of the display of query results.
- **METER** Stopwatch testing performed on 250 test queries that represent a defined usage operational profile.
- **MUST** No more than 10 seconds.
- **PLAN** No more than three seconds.
- **WISH** No more than two seconds for all queries.

